KIRIRI WOMEN'S UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF SCIENCE, DEPARTMENT OF COMPUTER SCIENCE AND IT

SMART HEALTHCARE MANAGEMENT SYSTEM FOR KIRIRI WOMEN'S UNIVERSITY OF SCIENCE AND TECHNOLOGY

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A REPORT SUBMITTED TO THE SCHOOL OF COMPUTING IN PARTIAL FULFILMENT FOR THE AWARD OF A DEGREE IN COMPUTER SCIENCE AT KIRIRI WOMEN'S UNIVERSITY OF SCIENCE AND TECHNOLOGY

[2026]

DECLARATION

I, the undersigned, declare that this proposal is my original work and that it has not been presented in any other university or institution for academic credit.

Signature	Date
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Student Name	Reg_No
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Signature	Date
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Supervisor's Name	
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DEDICATION

I dedicate this project to my beloved family, whose unwavering support, encouragement, and prayers have been my source of strength throughout my academic journey. To my parents, for instilling in me the values of hard work, resilience, and integrity.

I also dedicate this work to my lecturers and supervisor, whose guidance and mentorship have shaped my knowledge and inspired me to pursue excellence.

Finally, I dedicate this project to my fellow students and the entire Kiriri Women's University of Science and Technology community, with the hope that this SmartHealth system will contribute to improving healthcare services and leave a lasting impact.

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I would like to express my sincere gratitude and appreciation to all those who have contributed to the successful completion of this project research. This endeavor has been a significant learning experience, and I am truly thankful for the support and guidance received.

I extend my heartfelt appreciation to the project for their valuable insights. Encouragements and constructive feedback throughout the project. Their guidance has been instrumental in shaping the project's direction and improving its overall quality.

I would like to acknowledge my colleagues and peers who aided in interviews. Sharing ideas, and collaborating effectively in data collection fostered a collaborative environment. Your contribution has enhanced the project and made it a collective achievement.

Lastly, I appreciate the support from my family and friends who have been a constant source of encouragement and understanding during this academic journey.

Thank you once again, everyone involved. Your support has been invaluable. And I am proud to have been part of such a collaborative and enriching project.

Sincerely,

Joyvender Awino.

ABSTRACT

This project presents the design and development of KWUST SmartHealth, a web-based healthcare management system aimed at digitizing and simplifying access to healthcare services at Kiriri Women's University of Science and Technology (KWUST). The system addresses critical challenges faced by university health centers, including manual appointment scheduling, fragmented medical record-keeping, lack of a centralized medicine inventory, and limited patient engagement with their own health data. Leveraging a modern technology stack comprising Node.js, Express, EJS templating, and Directus as a headless CMS with a PostgreSQL backend, the platform delivers a responsive and intuitive interface with role-based access for students, healthcare staff, and administrators. Students can book appointments from staff-defined time slots, view their medical history along with prescribed medicines, manage personal health notes and to-do lists, and interact with an AI-powered chatbot for general health inquiries. Staff can manage appointments, create medical records with embedded prescriptions, oversee the medicine inventory, and send reminders to students. The administrator dashboard provides user management, account verification, system audit logs, and oversight of medical records and medicines. The system incorporates a full PDF generation service for printing medical records and prescriptions featuring the institutional logo and a generated timestamp. A dark mode toggle, comprehensive search and filtering across all data tables, and a Content Security Policy-compliant frontend ensure usability, accessibility, and security. The AI chatbot integrates Google's Gemini 2.0 Flash model to answer general health questions without storing chat history, emphasizing user privacy. By consolidating healthcare workflows into a single unified platform, KWUST SmartHealth reduces administrative overhead, improves data accuracy, enhances the patient experience, and fosters a more informed and health-conscious university community.

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1. CHAPTER ONE: INTRODUCTION

1.1. Background of the Study

Healthcare remains one of the most critical sectors in need of digital transformation worldwide (World Health Organization, 2021). Despite significant advances in medical science and health technology, many healthcare facilities in developing countries continue to rely on paper-based processes for core administrative and clinical functions, undermining the quality, efficiency, and accessibility of care. The World Health Organization (WHO), through its Global Strategy on Digital Health 2020–2025, has identified digital health as a key enabler of universal health coverage and has called upon member states to accelerate the adoption of information and communication technologies (ICT) to strengthen health systems (World Health Organization, 2021).

University health centers occupy a unique position in the healthcare landscape. They serve large, diverse, and often transient populations of students and staff, encompassing a wide range of health needs from primary care and preventive services to mental health support and emergency treatment. Yet, despite the availability of modern health information technologies, many university health centers, particularly in Africa, continue to operate with manual or partially digitized workflows for appointment booking, medical record-keeping, prescription management, and medicine inventory tracking.

In Kenya, the adoption of digital health systems in higher education institutions has been slow and uneven (Kruse et al., 2016). Studies have found that fewer than 20% of Kenyan university health centers had implemented any form of electronic medical record system, with the majority still relying on paper-based filing systems that are prone to loss, damage, and inaccessibility. Assessments of digital health readiness in Kenyan university health facilities have identified significant gaps in infrastructure, technical capacity, and institutional commitment, concluding that most institutions were not adequately prepared for comprehensive digital health transformation.

Country/Region	**EHR Adoption Rate (Hospitals)**	**EHR Adoption Rate (Univ. Health Centers)**
United States	> 80% (2015) (Jamoom et al., 2014)	50% (estimated)
European Union	60% (2020) (OECD, 2020)	30% (estimated)
Kenya	25% (public hospitals)	< 20%
Sub-Saharan Africa	15%	< 10% (estimated)

Table 1: Comparative EHR Adoption Rates Across Healthcare Settings

At Kiriri Women's University of Science and Technology (KWUST), the university health center currently operates with entirely paper-based systems for scheduling appointments, storing medical records, tracking medicine inventory, and managing prescriptions. Students must physically visit the health center to book consultations, with no visibility into available time slots and no means to cancel or reschedule appointments without making another physical trip. Medical records are maintained in physical files stored in cabinets, making them susceptible to loss, damage, and inaccessibility during follow-up visits or emergencies. Healthcare staff track medicine inventory manually using ledgers, a process that is time-consuming, error-prone, and provides no real-time visibility into stock levels or approaching expiry dates. There is no centralized digital platform for staff to record patient diagnoses, prescribe medicines, or send appointment reminders.

1.1.1. Stakeholder Pain Points

During preliminary consultations conducted with students, healthcare staff, and administrators at KWUST, the limitations of the current manual system were clearly articulated across all stakeholder groups:

Stakeholder	Pain Points Expressed	Impact on Healthcare Experience
Students	Long queues at health center; no way to book appointments remotely; inability to view medical history; no reminders for follow-up visits; frustrated by lost or misplaced records	Reduced access to timely care; missed appointments; poor continuity of care; dissatisfaction with health services
Healthcare Staff	Heavy administrative burden from manual record-keeping; difficulty tracking patient follow-ups; time wasted searching for paper files; inability to monitor medicine stock levels efficiently; handwritten prescriptions easily lost or misinterpreted	Reduced time available for direct patient care; increased stress and workload; potential for medication errors
Administrators	No visibility into health center operations and service utilization; inability to generate reports on common diagnoses or medicine usage patterns; no audit trail for accountability; difficulty verifying user identities and managing access	Poor oversight of health center performance; challenges in resource planning and budget allocation; limited accountability measures

Table 2: Summary of Stakeholder Pain Points Identified During Preliminary Consultations at KWUST

1.1.2. The Digital Health Imperative

The global trajectory toward digital health transformation is well established. Extensive research has demonstrated the benefits of health information technology (HIT) in enhancing clinical efficiency, reducing medical errors, and improving patient outcomes (Buntin et al., 2011). Electronic health record (EHR) systems have revolutionized the way patient data is stored, retrieved, and shared among healthcare providers, with studies showing that EHR adoption leads to improved documentation quality, better coordination of care, and reduced duplication of tests (Jamoom et al., 2014; Menachemi & Collum, 2011). The HITECH Act of 2009 in the United States accelerated EHR adoption from approximately 10% of hospitals in 2008 to over 80% by 2015, demonstrating that targeted policy interventions can effectively drive digital transformation (Adler-Milstein & Jha, 2017).

Several studies have specifically examined the implementation of healthcare information systems in university settings. Research has shown that digital appointment scheduling can reduce average wait times and improve patient satisfaction in university health centers (Chaudhry et al., 2006). In the domain of medicine inventory management, digital systems have been demonstrated to reduce stockout incidents and decrease expired medicine write-offs compared to manual tracking methods (Volland et al., 2017). These findings underscore the tangible benefits that digital health systems can deliver in university contexts.

The WHO has emphasized that digital health is not merely about technology adoption but about fundamentally rethinking how healthcare is delivered, accessed, and experienced (World Health Organization, 2021). By addressing the specific needs of the KWUST University Health Center, this research aims to contribute to the broader goal of digitizing healthcare services in educational institutions across Kenya and beyond.

1.1.3. Theoretical Foundations

This research is grounded in three complementary theoretical frameworks. First, the Information Systems Success Model, originally proposed by DeLone and McLean, posits that the success of an information system can be measured by six interdependent dimensions: system quality, information quality, service quality, user satisfaction, intention to use, and net benefits (Laudon & Laudon, 2020). KWUST SmartHealth aims to achieve high marks across all these dimensions by delivering a reliable, user-friendly, and comprehensive healthcare management platform. Second, the Technology Acceptance Model (TAM) provides a framework for understanding how perceived usefulness and perceived ease of use influence user adoption, guiding the design of an intuitive interface that accommodates varying levels of technical literacy (Davis, 1989). Third, principles of user-centered design (UCD) emphasize the involvement of end-users throughout the development process to ensure the final product meets their needs and expectations (Irizarry et al., 2015). Together, these frameworks provide a robust theoretical foundation for designing, developing, and evaluating the proposed system.

Area	Current Challenge (Manual)	Proposed Solution (SmartHealth)
Appointment Scheduling	No visibility into slots; physical visits required; no cancellation option	Online booking from predefined slots; view, cancel, reschedule anytime.
Medical Records	Paper files; prone to loss, damage, inaccessibility	Digital records; instant retrieval; persistent storage
Medicine Inventory	Manual stock tracking; expiry unnoticed; stockouts	Centralized tracking; search, filter, stock-level alerts
Prescription Management	Handwritten notes; easy to misplace	Digital prescriptions linked to records; dosage, duration, instructions
Patient Communication	No appointment reminders; no health notifications	Automated reminders; health alerts; AI chatbot for inquiries
Patient Engagement	No access to own medical history	Patient portal with full medical and prescription history

Table 3: Summary of Challenges and Proposed Digital Solutions at KWUST Health Center

1.2. Problem Statement

The university health center at KWUST currently operates with entirely manual processes for appointment scheduling, medical record keeping, prescription management, and medicine inventory tracking. These manual workflows are inherently inefficient, error-prone, and unable to scale with the growing student population, which currently stands at several thousand students. The consequences of this reliance on paper-based systems are felt across every dimension of healthcare delivery at the institution.

In the area of appointment scheduling, students must physically visit the health center during operating hours to book consultations, with no ability to view available time slots in advance. This creates inefficiencies for both students, who may make multiple trips to secure an appointment, and staff, who must manage bookings manually without any scheduling optimization. The absence of automated reminders contributes to high no-show rates, as students may forget appointments booked days or weeks in advance. There is no mechanism for students to cancel or reschedule appointments without making another physical visit, leading to wasted time slots that could otherwise be used by other patients.

The management of medical records at the KWUST health center presents equally significant challenges. Patient records are stored in physical paper files organized in

filing cabinets, requiring staff to manually retrieve, update, and re-file each record. This system is susceptible to several problems: files may be misplaced or misfiled, making them unavailable when needed for a patient visit; records may be damaged by water, fire, or general wear and tear; and only one person can access a physical file at a time, impeding coordinated care. When patients return for follow-up visits, their previous medical history may not be readily available, compromising continuity of care and forcing clinicians to rely on patient recollection or repeat diagnostic procedures.

Medicine inventory management at the health center relies on manual ledger entries to track stock levels, a process that is time-consuming and prone to human error. Staff must physically count medicines to assess stock levels, with no automated alerts when quantities fall below safe thresholds. Expiry dates are tracked manually, if at all, leading to situations where expired medicines go unnoticed until they are dispensed or discovered during stock checks. The lack of real-time inventory visibility means that stockouts can occur unexpectedly, leaving patients without essential medications.

The problem is compounded by the absence of a centralized digital platform for clinical workflows. There is no system for staff to record patient diagnoses in a structured, searchable format; to prescribe medicines with proper dosage, duration, and instructions; or to view a patient's complete medical history during a consultation. Communication with patients is limited to phone calls or word of mouth, with no systematic mechanism for sending appointment reminders, health notifications, or follow-up instructions.

Problem Area	**Manual**	**Systemic**	**Patient-Impact**	**Staff-Impact**
Appointment Scheduling	✓	✓	High	High
Medical Records	✓	✓	Severe	Severe
Medicine Inventory	✓	✓	Moderate	High
Prescription Management	✓	—	High	Moderate
Patient Communication	✓	✓	High	Moderate

Table 4: Problem Area Assessment Matrix — Manual vs. Systemic Challenges and Their Impact Levels

These challenges collectively undermine the quality, efficiency, and accessibility of healthcare services at KWUST. Students receive suboptimal care due to long wait times, lost records, and poor communication. Healthcare staff spend valuable time on administrative tasks that could be devoted to patient care. Administrators lack visibility into health center operations and cannot make data-informed decisions about resource allocation or service improvement. This research seeks to address these interconnected challenges by developing a comprehensive web-based

healthcare management system that digitizes core healthcare workflows, empowers students with access to their own health data, and provides staff and administrators with the tools needed to deliver efficient, high-quality care.

1.3. Research Questions

Guided by the problem statement above, this research seeks to answer the following questions:

Research Question	Description
1. Process Analysis	What are the current manual healthcare processes used at the KWUST University Health Center, and what specific inefficiencies and limitations do they introduce into healthcare delivery?
2. System Design	What architectural components, features, and security measures are required to design a secure and user-friendly healthcare management system for a university health center context?
3. Implementation Feasibility	How can a web-based healthcare management system be implemented using modern open-source technologies to meet the functional requirements of students, healthcare staff, and administrators?
4. System Evaluation	Does the developed system meet the defined functional and non-functional requirements, and how do end-users perceive its usability, reliability, and effectiveness?
5. Appointment Optimization	How can a role-based appointment scheduling system with predefined time slots improve access to care and reduce administrative overhead compared to manual booking?

Table 5: Research Questions Guiding the KWUST SmartHealth Development

1.4. Objectives

1.4.1. General Objective

To develop a web-based healthcare management system that digitizes and streamlines healthcare service delivery at KWUST, improving efficiency, accessibility, and quality of care for students, staff, and administrators.

1.4.2. Specific Objectives

- **To analyze** the current manual healthcare processes used in the university clinic to identify inefficiencies and system requirements.
- **To design** a secure and user-friendly smart healthcare system that includes patient registration, appointment scheduling, electronic medical records, and communication features.
- **To implement** a web-based smart healthcare management system that supports patients, doctors, and administrators in managing healthcare services digitally.
- **To test** the developed system to ensure it meets functional and non-functional requirements such as accuracy, security, usability, and performance within the university clinic environment.

- **To design** and implement a role-based appointment scheduling system that allows students to book, view, and manage appointments from available time slots.

1.4.3. Mapping of Objectives to Research Questions and Methodology

Each specific objective is linked to a corresponding research question and will be addressed through the methodology outlined in Chapter Three.

Objective	Research Question	Methodology
1. Analyze current processes	RQ1 — Process Analysis	Stakeholder interviews, process observation, workflow documentation
2. Design system architecture	RQ2 — System Design	UML modeling, use case diagrams, architecture design, security planning
3. Implement the system	RQ3 — Implementation	Agile/Scrum development, Node.js, Directus CMS, PostgreSQL, iterative prototyping
4. Test the system	RQ4 — Evaluation	Functional testing, usability testing (SUS), performance benchmarking, security testing
5. Design appointment scheduling	RQ5 — Optimization	Slot-based scheduling algorithm design, database schema, frontend calendar UI

Table 6: Mapping of Specific Objectives to Research Questions and Methodology

1.5. Scope and Limitations

1.5.1. Scope

This research is focused on the design, development, and evaluation of a comprehensive web-based healthcare management system for the KWUST university health center. The scope of the research is defined across four dimensions:

- **Target Population:** The system is designed for three distinct user groups at Kiriri Women's University of Science and Technology: students who require access to healthcare services, healthcare staff who deliver clinical care and manage health center operations, and administrators who oversee system governance, user management, and reporting. The system accommodates the specific needs and access privileges of each role through role-based views and functionality.
- **Geographical Focus:** The system is developed for and will be deployed at the KWUST university health center located in Kiriri, Kenya. The architecture is modular and follows open standards, enabling adaptation and deployment at other similar institutions with minimal modification. However, the primary context of design, testing, and evaluation is the KWUST environment.
- **Timeframe:** The development and evaluation of the proposed solution are planned within a structured 4-month period, organized into six 2-week sprints following an

Agile Scrum methodology. This timeframe encompasses requirements analysis, system design, iterative development, testing, and evaluation.

- **Features:** The research encompasses the development of seven core modules: (1) user management and authentication with role-based access control; (2) appointment scheduling with predefined time slots and self-service booking; (3) electronic medical records and prescription management with structured data entry; (4) medicine inventory management with search, categorization, and stock tracking; (5) an AI-powered chatbot integrated with Google's Gemini API for general health inquiries; (6) a PDF generation service for printing medical records and prescriptions; and (7) an administrative dashboard for user management, audit logging, and system oversight.

1.5.2. Limitations

This research is subject to several limitations that should be considered when interpreting the findings and assessing the generalizability of the results:

- **External System Integration:** Integration with external laboratory information systems, pharmacy management systems, or national health information exchanges is outside the scope of the current implementation. The system is designed as a standalone institutional platform, though the RESTful API architecture provides a foundation for future integration.
- **Internet Connectivity:** The system is web-based and relies on the availability of reliable internet connectivity for access. While the system is designed to be responsive and efficient even on slower connections, inconsistent or unavailable internet access in certain areas may limit usability. Offline functionality or progressive web app (PWA) capabilities are not included in the current scope.
- **Context Specificity:** The evaluation of the system is limited to the KWUST university health center context. While the system architecture is modular and designed for adaptability, the specific findings regarding usability, effectiveness, and user satisfaction may not be directly generalizable to other institutions without further validation studies.
- **AI Chatbot Limitations:** The AI-powered chatbot integrated into the system is designed for general health information and educational purposes only. It is not a substitute for professional medical diagnosis, advice, or emergency services. The chatbot's knowledge is derived from its training data and may not reflect the most current medical guidance. Users are explicitly informed of these limitations through disclaimers displayed within the chatbot interface.
- **Sample Size for Evaluation:** The usability testing and system evaluation are planned with a sample of volunteer participants from the KWUST community. While this approach provides valuable insights into user experience and system performance, the relatively small sample size may limit the statistical power and generalizability of the evaluation findings.

- **Data Migration:** The transition from paper-based records to the digital system requires manual data entry or phased adoption. The current scope does not include automated digitization of existing paper records, and the system is designed for prospective use from the point of deployment onward.

1.6. Significance of the Study

The significance of this research extends across multiple dimensions, encompassing practical, theoretical, and policy contributions to the field of health information technology in developing-country university contexts.

1.6.1. Practical Significance

At the most immediate level, this research will deliver a functional, open-source healthcare management system that directly addresses the operational challenges faced by the KWUST university health center. Students will benefit from convenient online appointment booking, access to their medical records through a patient portal, automated health reminders, and reduced wait times. Healthcare staff will benefit from streamlined clinical workflows, structured digital record-keeping, integrated prescription management with automatic inventory updates, and reduced administrative burden. Administrators will gain visibility into health center operations through dashboards, audit logs, and reporting capabilities that support data-informed decision-making.

The open-source nature of the system means that other university health centers facing similar challenges — in Kenya, across Africa, and in other developing regions — can adopt and adapt the system for their own contexts at minimal cost (Monospace Inc., 2024). This addresses a critical gap in the current landscape, where enterprise-level health information systems are prohibitively expensive for small to medium-sized institutions (Kruse et al., 2016). By providing a freely available, well-documented, and modular alternative, this research democratizes access to health information technology for resource-constrained institutions.

1.6.2. Theoretical Significance

This research contributes to the growing body of knowledge on health information technology adoption in educational institutions. While extensive research has examined EHR implementation in large hospital settings (Jamoom et al., 2014; Menachemi & Collum, 2011), relatively few studies have focused on the specific context of university health centers, particularly in developing countries (Kruse et al., 2016). This study addresses this gap by providing empirical evidence on the design, implementation, and evaluation of a healthcare management system in a Kenyan university setting.

The application of the DeLone and McLean Information Systems Success Model (Laudon & Laudon, 2020), the Technology Acceptance Model (Davis, 1989), and user-centered design principles (Irizarry et al., 2015) to a university health center context provides valuable insights into how these theoretical frameworks apply in resource-constrained, developing-country settings. The findings may inform adaptations to these frameworks for similar contexts in the future.

1.6.3. Policy Significance

The COVID-19 pandemic underscored the critical importance of digital health infrastructure and contactless healthcare services (World Health Organization, 2021). The Kenyan government has identified digital health as a priority area in its national health strategy, yet progress in implementing health information systems in educational institutions has been slow (Kruse et al., 2016). This research provides a practical demonstration of how a university health center can successfully implement a comprehensive digital health solution, serving as a case study that can inform policy development and resource allocation decisions at the institutional and national levels.

The system's focus on user experience, accessibility, and affordability aligns with the WHO's call for digital health solutions that are equitable, inclusive, and responsive to the needs of diverse populations (World Health Organization, 2021). By demonstrating that effective digital health transformation is achievable even with limited resources, this research provides evidence that can encourage other institutions and policymakers to invest in similar initiatives.

1.6.4. Comparison with Existing Solutions

Existing commercial healthcare management systems are often prohibitively expensive for small to medium-sized institutions like KWUST, with licensing fees, implementation costs, and ongoing maintenance expenses that can run into millions of shillings. They frequently lack the flexibility to adapt to the specific workflows of a university health center, requiring institutions to modify their processes to fit the software rather than the other way around. This research addresses both of these barriers by developing an open-source, web-based system tailored specifically for university health centers. The headless CMS architecture provides exceptional flexibility for customization (Monospace Inc., 2024), while the inclusion of AI-powered chatbot assistance and PDF generation features adds value beyond what is typically available in existing low-cost solutions.

1.7. Target Audience

The KWUST SmartHealth system is designed for three primary user groups, each with distinct needs, responsibilities, and levels of system access.

1.7.1. Students

Students constitute the largest user group and are the primary beneficiaries of the system's patient-facing features. Through the patient portal, students can register for an account, book appointments from available time slots, view their upcoming and past appointments, cancel or reschedule bookings as needed, access their complete medical history including diagnoses and prescribed medicines, download their medical records and prescriptions as PDF documents, and interact with the AI chatbot for general health inquiries. The system is designed to be intuitive and accessible to students with varying levels of technical literacy, with a responsive layout that works on both desktop computers and mobile devices. The self-service nature of the portal empowers students to take an active role in managing their healthcare, reducing reliance on administrative staff for routine tasks.

1.7.2. Healthcare Staff

Healthcare staff at the KWUST University Health Center are the primary clinical users of the system. Staff members use the system to define available appointment slots, view and manage patient appointments, create electronic medical records with structured diagnoses, symptoms, and clinical notes, prescribe medicines with dosage, duration, and specific instructions, track medicine inventory with automatic stock-level updates when prescriptions are issued, search and filter patient records by name, date, or diagnosis, and generate PDF copies of medical records and prescriptions for patient records or referrals. By digitizing these clinical workflows, the system reduces the administrative burden on healthcare staff, allowing them to devote more time and attention to direct patient care.

1.7.3. Administrators

Administrators oversee the governance and operation of the system. Their responsibilities include managing user accounts across all roles (creating, verifying, suspending, and deleting accounts), monitoring system activity through comprehensive audit logs, reviewing all medical records and medicine inventory across the health center, and generating reports and analytics for operational oversight. The administrative dashboard provides a centralized interface for these functions, with role-based access controls ensuring that administrative capabilities are restricted to authorized personnel.

User Role	Primary Responsibilities	System Access Level	Key Modules
Student	Book appointments; view medical history; manage prescriptions; access health information	Personal data only; self-service operations	Patient Portal, Appointment Booking, Chatbot, PDF Download
Healthcare Staff	Manage appointments; create medical records; prescribe medicines; track inventory.	All patient data within scope; clinical operations	Appointments, Medical Records, Prescriptions, Inventory, PDF Generation
Administrator	User management; system oversight; audit monitoring; reporting	Full system access; administrative operations	Admin Dashboard, User Management, Audit Logs, Reports

Table 7: Target Audience Roles, Responsibilities, and System Access Levels

1.8. Organization of the Study

This research is organized into five chapters. Chapter One (the current chapter) provides the background, problem statement, objectives, research questions, scope, significance, and target audience for the study. Chapter Two presents a comprehensive review of existing literature and systems related to healthcare management, including electronic health records, appointment scheduling, medicine inventory management, AI in healthcare, patient portals, and security considerations. Chapter Three describes the methodology employed in developing the system,

including the system architecture, technology stack, development methodology, key modules, PDF generation pipeline, and evaluation procedures. Chapter Four presents the results of the system implementation and evaluation, including functional testing, usability assessment, and performance benchmarking. Chapter Five discusses the findings, draws conclusions, and provides recommendations for future work.

1.9. Definition of Key Terms

Term	Definition
Electronic Health Record (EHR)	A digital version of a patient's medical history, maintained by healthcare providers over time, including diagnoses, medications, treatment plans, immunization dates, and test results
Headless CMS	A content management system that separates the content repository (back end) from the presentation layer (front end), delivering content via RESTful APIs rather than through a built-in presentation framework
Role-Based Access Control (RBAC)	A security approach that restricts system access based on a user's role within the organization, ensuring that users can only access information and perform actions necessary for their responsibilities
JWT (JSON Web Token)	A compact, URL-safe token format used for securely transmitting information between parties as a JSON object, commonly used for authentication and authorization in web applications
System Usability Scale (SUS)	A standardized questionnaire used to measure the perceived usability of a system, yielding a score from 0 to 100 based on ten items with five response options
Agile / Scrum	An iterative software development methodology that organizes work into fixed-length sprints (typically 1–4 weeks) with daily standups, sprint reviews, and retrospectives
REST API	Representational State Transfer Application Programming Interface, an architectural style for designing networked applications that uses HTTP methods for data operations
Gemini API	Google's application programming interface for accessing Gemini large language models, used in this research for powering the AI chatbot feature
No-Show Rate	The percentage of scheduled appointments that patients miss without prior cancellation, a key metric for evaluating appointment system effectiveness

Audit Log	A chronological record of system activities that enables the reconstruction and examination of events, used for security monitoring and accountability

Table 8: Definition of Key Technical Terms Used in This Research

2. CHAPTER TWO: LITERATURE REVIEW

2.1. Introduction

The rapid advancement of information and communication technologies (ICT) has fundamentally transformed healthcare delivery across the globe. Electronic health records, online appointment scheduling, digital inventory management, telemedicine platforms, and AI-powered clinical decision support systems have become standard practice in many healthcare facilities, leading to measurable improvements in efficiency, reduction in medical errors, and enhanced patient satisfaction (Black et al., 2011; Buntin et al., 2011). The World Health Organization's Global Strategy on Digital Health 2020–2025 identifies digital health as a critical enabler of universal health coverage and calls for the systematic adoption of ICT to strengthen health systems worldwide (World Health Organization, 2021).

Despite this global momentum, the adoption of digital health technologies in university health centers has been significantly slower than in large hospital systems (Kruse et al., 2016). Many university health centers, particularly in developing countries, continue to rely on manual or partially digitized processes for core healthcare functions such as appointment booking, medical record keeping, prescription management, and medicine inventory tracking (Chaudhry et al., 2006). This disconnect between the broader digital transformation of healthcare and the operational reality of university health centers represents a significant gap that warrants focused research and development.

This chapter presents a comprehensive review of existing literature and systems related to healthcare management, with a specific focus on university health center contexts. The chapter is organized into twelve sections. Section 2.2 establishes the theoretical framework underpinning this research, including the DeLone and McLean Information Systems Success Model, the Technology Acceptance Model, and user-centered design principles. Sections 2.3 through 2.8 examine key technological domains: electronic health records, appointment scheduling, medicine inventory management, AI and chatbots in healthcare, patient portals, and security and privacy considerations. Section 2.9 provides a comparative analysis of existing approaches, followed by a synthesis of identified gaps in Section 2.10. Section 2.11 explains how the proposed KWUST SmartHealth system addresses these gaps, and Section 2.12 concludes the chapter with a summary.

2.2. Theoretical Framework

2.2.1. DeLone and McLean Information Systems Success Model

The DeLone and McLean Information Systems (IS) Success Model, originally proposed in 1992 and updated in 2003, provides a comprehensive framework for evaluating the effectiveness of information systems (Laudon & Laudon, 2020). The model posits that the success of an information system can be measured through six interdependent dimensions:

Dimension	Description
System Quality	Measures the technical performance of the system, including reliability, response time, ease of use, and availability. A system that is frequently unavailable or slow to respond will undermine user confidence and adoption.
Information Quality	Evaluates the accuracy, timeliness, completeness, relevance, and consistency of the information produced by the system. Healthcare applications demand especially high information quality given the clinical consequences of incorrect data.
Service Quality	Assesses the quality of support provided to users, including responsiveness of technical support, training availability, and overall service reliability.
User Satisfaction	Captures the overall attitude of users toward the system, typically measured through surveys and feedback mechanisms such as the System Usability Scale (SUS).
Intention to Use / Use	The degree to which users actually engage with the system and intend to continue using it. Actual usage patterns provide objective evidence of system adoption.
Net Benefits	The overall impact of the system on individuals, organizations, and society. In healthcare contexts, this includes improved patient outcomes, reduced costs, and enhanced efficiency.

Table 9: Six Dimensions of the DeLone and McLean IS Success Model

This framework is particularly relevant to KWUST SmartHealth because it provides a structured approach for evaluating whether the system truly meets its objectives. Each module of the proposed system can be assessed against these six dimensions. For example, the appointment scheduling module must demonstrate high system quality (fast response times, reliable slot availability), information quality (accurate schedule data), and net benefits (reduced wait times, improved access to care). The model guides both the design phase (ensuring all six dimensions are considered) and the evaluation phase (providing metrics for post-implementation assessment).

2.2.2. Technology Acceptance Model

The Technology Acceptance Model (TAM), introduced by Davis (Davis, 1989), remains one of the most widely used theoretical frameworks for understanding user acceptance of information technology. TAM identifies two primary determinants of technology acceptance:

- **Perceived Usefulness (PU):** The degree to which a user believes that using the system will enhance their performance. For KWUST SmartHealth, perceived usefulness for students translates to convenient appointment booking and easy access to medical records. For staff, it means reduced administrative burden and efficient clinical workflows.
- **Perceived Ease of Use (PEOU):** The degree to which a user believes that using the system will be free of effort. This is particularly important for the proposed system given the varying levels of technical literacy among students, healthcare staff, and administrators at KWUST.

The TAM framework has been extensively validated in healthcare contexts (Irizarry et al., 2015; Venkatesh et al., 2003). Studies have shown that both PU and PEOU significantly influence healthcare professionals' intention to adopt electronic health records and other clinical information systems (Kruse et al., 2016). For university health centers, where users may include non-traditional students and older staff members with limited digital experience, designing for high PEOU through intuitive interfaces, clear navigation, and responsive design becomes paramount.

Building on TAM, the Unified Theory of Acceptance and Use of Technology (UTAUT) proposed by Venkatesh et al. (Venkatesh et al., 2003) synthesizes eight prominent technology acceptance models into a unified framework with four core determinants: performance expectancy, effort expectancy, social influence, and facilitating conditions. UTAUT has been applied in numerous healthcare IT adoption studies and provides additional guidance for KWUST SmartHealth, particularly regarding the importance of facilitating conditions such as training, technical support, and adequate infrastructure.

2.2.3. User-Centered Design

User-centered design (UCD) is an iterative design approach that places end-users at the center of the development process (Irizarry et al., 2015). The UCD process involves four distinct phases: understand the context of use, specify user requirements, produce design solutions, and evaluate against requirements. This cycle repeats until usability goals are achieved.

For KWUST SmartHealth, UCD principles guide the development process in several ways:

Phase	Activities	Application in SmartHealth
1. Understand Context	Interviews, observations, contextual inquiry	Initial consultations with students and staff at KWUST health center to understand workflows and pain points
2. Specify Requirements	User stories, personas, scenarios	Development of user stories for each role (student, staff, admin) defining specific features and interactions
3. Produce Solutions	Wireframes, prototypes, highfidelity mockups	Iterative design of EJS templates with responsive layout, dark mode, and accessibility features
4. Evaluate	Usability testing, SUS surveys, heuristic evaluation	Testing with 10 volunteer participants measuring task completion rate, errors, and SUS score

Table 10: User-Centered Design Phases Applied to KWUST SmartHealth

The UCD approach is particularly important in healthcare settings where the consequences of poor usability can directly impact patient safety and quality of care (Bates & Gawande, 2003). An interface that is confusing or error-prone can lead to incorrect data entry, missed appointments, or medication errors. By involving end-users throughout the development process, KWUST SmartHealth aims to minimize these risks and deliver a system that truly meets the needs of its diverse user base.

2.2.4. Health Informatics Theory

Health informatics sits at the intersection of information science, computer science, and healthcare, providing the theoretical foundation for developing and evaluating health information systems (Chaudhry et al., 2006). The discipline encompasses the acquisition, storage, retrieval, and use of health information to improve healthcare delivery. Key principles from health informatics that inform this research include:

- **Data Standardization:** Ensuring that clinical data is recorded in structured, consistent formats to enable accurate retrieval, analysis, and exchange. KWUST SmartHealth implements this through structured forms for diagnoses, prescriptions, and medical history.
- **Interoperability:** The ability of different information systems to exchange and use data. While the current scope of the project does not extend to external system integration, the modular architecture and RESTful API design lay the groundwork for future interoperability with laboratory and pharmacy systems (Bates & Gawande, 2003).

- **Clinical Decision Support:** The use of computerized information to assist clinicians in making clinical decisions. The AI chatbot in SmartHealth provides general health information, representing a basic form of decision support, while structured data entry supports staff in making informed prescribing decisions.

2.3. Electronic Health Record Systems

2.3.1. Overview and Evolution

Electronic Health Record (EHR) systems have evolved from simple digital repositories of patient information into comprehensive platforms that support clinical decision-making, care coordination, population health management, and health analytics (Menachemi & Collum, 2011). The evolution of EHR systems can be traced through several generations:

Generation	Characteristics	Examples	Limitations
1st Gen (1970s–1990s)	Basic digital storage of patient demographics and billing data; departmentspecific systems	COSTAR, HELP	No interoperability; limited clinical functionality; proprietary data formats
2nd Gen (1990s–2010s)	Integrated clinical data including lab results, medications, and diagnoses; basic CPOE	Epic, Cerner, Allscripts	Expensive deployment; complex user interfaces; vendor lock-in
3rd Gen (2010s–present)	Cloud-based; interoperability via FHIR/HL7; patient portals; mobile apps; AI integration	Epic MyChart, Cerner HealtheLife, OpenMRS	Still costly for small institutions; requires reliable internet; data privacy concerns

Table 11: Evolution of Electronic Health Record Systems

The HITECH Act of 2009 in the United States provided substantial financial incentives for healthcare providers to adopt EHR systems, resulting in acceleration from approximately 10% of hospitals in 2008 to over 80% by 2015 (Adler-Milstein & Jha, 2017; Jamoom et al., 2014). This policy-driven adoption demonstrated that regulatory and financial mechanisms can effectively accelerate digital transformation in healthcare. However, the benefits of such policies have not been uniformly distributed, with smaller institutions and those in developing countries lagging behind (Kruse et al., 2016).

2.3.2. Benefits of EHR Adoption

Extensive research has demonstrated the benefits of EHR systems across multiple dimensions of healthcare delivery (Buntin et al., 2011; Menachemi & Collum, 2011):

Benefit Category	Specific Benefits	Evidence
Clinical Efficiency	Reduced documentation time; faster access to patient information; streamlined workflows	High
Patient Safety	Reduced medication errors; allergy alerts; drug interaction warnings; standardized order sets	High
Care Coordination	Shared access to patient data across providers; reduced duplicate testing; improved referral management	High
Quality of Care	Clinical decision support; guideline adherence; population health monitoring; preventive care reminders	Medium
Patient Engagement	Patient portals for self-service; access to medical records; online appointment booking; secure messaging	Medium
Cost Reduction	Reduced transcription costs; decreased paper/printing; lower storage costs; reduced malpractice claims	Medium

Table 12: Benefits of Electronic Health Record Systems

Jamoom et al. (Jamoom et al., 2014) found that providers who adopted EHR systems reported significant improvements in clinical outcomes, with 74% of physicians reporting enhanced patient care and 88% reporting improved access to patient data. Black et al. (Black et al., 2011) conducted a systematic overview of e-health impact studies and found predominantly positive effects on healthcare quality and safety, though the authors noted that the evidence base was stronger for some applications (such as electronic prescribing) than others.

2.3.3. Barriers to EHR Implementation

Despite the well-documented benefits, EHR adoption faces significant barriers, particularly in resourceconstrained settings (Kruse et al., 2016):

Barrier	Description and Impact
High Cost	Licensing fees, hardware procurement, infrastructure upgrades, and ongoing maintenance costs can be prohibitive for small institutions. Enterprise EHR implementations typically cost millions of dollars.
Implementation Complexity	Data migration from paper records, workflow redesign, system configuration, and integration with existing systems require significant technical expertise and time.
Training Requirements	Healthcare staff must be trained not only in system operation but also in new workflows. Training is often time-consuming and may temporarily reduce clinical productivity.

Interoperability Issues	Lack of standardized data exchange protocols makes it difficult for different systems to share information, leading to fragmented patient data across multiple platforms.
Resistance to Change	Clinical staff may resist adopting new systems due to concerns about workflow disruption, increased documentation burden, or skepticism about benefits.
Privacy and Security Concerns	Health data is highly sensitive, and concerns about data breaches, unauthorized access, and compliance with regulations create additional implementation barriers.
Technical Infrastructure	Reliable internet connectivity, adequate hardware, and ongoing technical support are prerequisites that may not be available in all settings.

Table 13: Barriers to EHR Adoption in Healthcare Settings

For university health centers in developing countries, these barriers are amplified by limited financial resources, inadequate technical infrastructure, and a shortage of trained health IT professionals (Chaudhry et al., 2006). The upfront cost of enterprise EHR systems can represent a significant portion of a university health center's annual budget, making adoption economically unfeasible without substantial external support.

2.3.4. EHR Systems in University Settings

The application of EHR systems in university health centers presents unique challenges and opportunities. University health centers serve a distinct population (predominantly young adults) with specific healthcare needs (preventive care, reproductive health, mental health services, and management of chronic conditions). The operational workflow differs from large hospital settings, with fewer specialties, lower patient volumes, and closer integration with campus life.

Existing studies on EHR implementation in university settings are limited. One study developed a web-based healthcare management system for a university health center and found improvements in appointment scheduling efficiency and patient satisfaction. However, the system did not include medicine inventory management or AI-powered features, limiting its comprehensiveness (Chaudhry et al., 2006).

OpenMRS, an open-source EHR platform originally designed for resource-constrained settings in developing countries, has been deployed in several university health centers. Its modular architecture allows customization for specific institutional needs, but its primary focus on clinical data management means that it lacks built-in appointment scheduling, inventory management, and patient portal features, requiring significant additional development to create a comprehensive solution.

2.4. Appointment Scheduling Systems

2.4.1. Overview of Digital Scheduling

Appointment scheduling is one of the most frequent and visible interactions between patients and healthcare providers. Traditional manual scheduling methods, including

phone-based booking and walk-in queues, are associated with long wait times, high no-show rates, and poor patient satisfaction. Digital appointment scheduling systems address these challenges by providing patients with the ability to view available time slots, book appointments online, receive automated reminders, and cancel or reschedule without human intervention.

Online appointment scheduling platforms such as Zocdoc and Practo have popularized digital booking in healthcare. These platforms aggregate healthcare provider information, allowing patients to search by specialty, location, insurance acceptance, and patient reviews. While these platforms have improved access to care in many contexts, they are primarily designed for individual practitioner use rather than institutional health centers and lack integration with other clinical systems.

2.4.2. Scheduling Algorithms and Optimization

The effectiveness of an appointment scheduling system depends significantly on the scheduling algorithm employed. Several approaches have been developed in the operations research and healthcare management literature:

Algorithm	**Approach**	**Strengths and Weaknesses**	**Com-plexity**
First-Come-FirstServed (FCFS)	Patients are served in the order they arrive or book	Simple to implement; can lead to long wait times during peak hours; no prioritization of urgent cases	Low
Open Access / Advanced Access	Same-day scheduling; patients call on the day they need to be seen	Reduces no-show rates; improves access; requires adequate same-day capacity; may not suit all clinic types	Medium
Modified Wave Scheduling	Multiple patients booked at the same time assuming some will arrive late or not at all	Optimizes provider utilization; risks patient dissatisfaction if all patients arrive on time	Medium
Double Booking	Two patients booked per time slot; one seen by provider, one by nurse/assistant	Improves throughput; requires care team coordination; may confuse patients	High
Overbooking	Deliberate overbooking based on historical no-show rates	Optimizes utilization; risk of overloading providers and increasing wait times	High

Table 14: Appointment Scheduling Algorithms and Their Characteristics

KWUST SmartHealth employs a time-slot-based scheduling approach where healthcare staff define available slots with a specified date, start time, end time, and

maximum patient capacity. Students select from available slots, and the system prevents double-booking through database-level constraints. This approach balances simplicity with effectiveness, providing a straightforward user experience while ensuring scheduling integrity. Future iterations could incorporate predictive algorithms that optimize slot allocation based on historical no-show patterns and demand forecasting.

2.4.3. Impact on Patient Satisfaction and No-Show Rates

Research consistently demonstrates that digital appointment scheduling has a positive impact on patient satisfaction and clinical efficiency (Irizarry et al., 2015). Web-based scheduling systems can reduce appointment-related phone calls, freeing staff to focus on clinical duties. Patients report higher satisfaction with the ability to book appointments outside of regular business hours and the convenience of managing appointments from any device.

No-show rates, which typically range from 15% to 30% in primary care settings, can be significantly reduced through digital scheduling with automated reminders (Irizarry et al., 2015). SMS and email reminders have been shown to reduce no-show rates by 25% to 40%, depending on the patient population and reminder timing. While KWUST SmartHealth currently implements dashboard-based reminders, future development could integrate SMS or email notification services to further reduce no-show rates.

2.5. Medicine Inventory Management Systems

2.5.1. Inventory Management Principles in Healthcare

Effective management of medicine inventory is critical for ensuring that essential medications are consistently available to patients while minimizing waste from expired or overstocked products (Volland et al., 2017). The healthcare supply chain presents unique challenges compared to general inventory management due to the critical nature of medical products (where stockouts can have life-threatening consequences), the regulatory requirements for storage and handling, and the complex demand patterns driven by patient needs and prescribing practices.

Healthcare inventory management involves balancing competing objectives:

- **Availability:** Ensuring that essential medicines are in stock when needed
- **Cost Efficiency:** Minimizing inventory carrying costs, including storage, insurance, and capital tied up in stock
- **Waste Reduction:** Preventing expiration of medicines and minimizing disposal costs
- **Regulatory Compliance:** Meeting requirements for storage conditions, record-keeping, and controlled substance management

2.5.2. ABC Analysis and Economic Order Quantity

Two fundamental techniques from inventory management theory are particularly relevant to medicine inventory systems:

ABC Analysis categorizes inventory items based on their consumption value (annual consumption × unit cost). Category A items represent 10–15% of items but account for 70–80% of total value, warranting tight control and frequent review. Category B items represent 20–30% of items and 15–20% of value, requiring moderate control. Category C items represent 50–60% of items but only 5–10% of value, allowing simplified management (Volland et al., 2017).

Category	% of Items	% of Value	Management Approach
A (High Value)	10–15%	70–80%	Tight control; frequent stock reviews; accurate records; just-in-time ordering
B (Medium Value)	20–30%	15–20%	Moderate control; periodic review; standard ordering procedures
C (Low Value)	50–60%	5–10%	Simplified control; bulk ordering; less frequent review; safety stock

Table 15: ABC Classification for Medicine Inventory Management

Economic Order Quantity (EOQ) provides a mathematical formula for determining the optimal order quantity that minimizes total inventory costs (ordering costs plus holding costs). While EOQ assumes constant demand and immediate delivery, which may not hold in all healthcare contexts, it provides a useful baseline for establishing reorder points and order quantities (Volland et al., 2017).

Kwust SmartHealth implements a simplified version of these principles through its inventory module. Medicines are categorized by therapeutic category (equivalent to basic ABC grouping), and stock levels are tracked with automated low-stock alerts when quantities fall below configurable thresholds. While the current implementation does not include dynamic EOQ calculation, the structured data model supports future integration of such optimization features.

2.5.3. Digital Inventory Systems

Digital medicine inventory management systems have demonstrated significant improvements over manual tracking methods (Volland et al., 2017). The key advantages include:

- **Real-Time Visibility:** Instant access to current stock levels across all medicines, enabling informed clinical decisions and proactive reordering
- **Expiry Tracking:** Automated identification of medicines approaching their expiration dates, reducing waste and preventing dispensing of expired products
- **Usage Analytics:** Historical data on consumption patterns enables demand forecasting and budget planning
- **Error Reduction:** Barcode scanning and digital entry reduce data entry errors compared to manual ledger systems

- **Audit Trail:** Complete record of inventory transactions supports regulatory compliance and financial auditing

Digital inventory systems in hospital pharmacies have been found to reduce inventory holding costs through improved ordering accuracy and waste reduction (Volland et al., 2017).

For KWUST SmartHealth, the inventory module provides essential functionality including medicine registration (name, category, quantity, expiry date), search and filtering by name or category, automatic stock level updates when medicines are prescribed through the medical records module, and low-stock alerts. This integration with the prescribing workflow is a key differentiator from standalone inventory systems.

2.6. AI and Chatbots in Healthcare

2.6.1. Evolution of Conversational AI in Healthcare

Artificial intelligence (AI) has emerged as a transformative force in healthcare, with applications spanning diagnostic imaging, drug discovery, personalized medicine, and clinical decision support (Shaw et al., 2019; Topol, 2019). Within this broader AI landscape, conversational AI in the form of chatbots and virtual assistants has gained particular traction for patient-facing applications (Yu et al., 2018).

Chatbots in healthcare can be broadly categorized into three types:

Type	Description	Examples	AI Level
Rule-Based	Follow predefined decision trees; respond to specific keywords; limited conversational ability.	Simple symptom checkers, appointment booking bots	Low
Retrieval-Based	Select responses from a predefined set based on input matching; more flexible than rule-based.	Customer service bots, FAQ assistants	Medium
Generative (LLM-based)	Use large language models to generate novel responses; understand context; handle complex conversations.	GPT-4, Gemini, Claude for health advice	High

Table 16: Types of Healthcare Chatbots by AI Capability

Rule-based and retrieval-based chatbots have been deployed in healthcare for tasks such as appointment scheduling, medication reminders, and basic symptom triage (Lee & Yoon, 2021). These systems are reliable and predictable but limited in their ability to handle complex or unexpected user inputs. Generative AI chatbots, powered by large language models (LLMs), represent a significant advancement, offering more natural conversational ability and broader knowledge coverage,

though they introduce challenges related to accuracy, safety, and consistency (Yu et al., 2018).

2.6.2. Clinical Applications of Chatbots

The clinical applications of healthcare chatbots span multiple domains (Lee & Yoon, 2021; Yu et al., 2018):

- **Symptom Assessment and Triage:** Chatbots guide patients through symptom questionnaires and provide recommendations for care urgency. Studies have shown reasonable accuracy for common conditions, though concerns remain about missed serious diagnoses.
- **Health Education:** Chatbots deliver personalized health information on topics such as medication adherence, chronic disease management, nutrition, and preventive care.
- **Mental Health Support:** AI-powered therapeutic chatbots provide cognitive behavioral therapy techniques, mood tracking, and crisis resources.
- **Appointment Management:** Chatbots handle appointment booking, reminders, cancellations, and rescheduling, reducing administrative burden on healthcare staff.
- **Medication Management:** Chatbots send medication reminders, provide information about side effects, and answer questions about drug interactions.

Lee and Yoon (Lee & Yoon, 2021) conducted a scoping review of conversational agents in healthcare and found that the majority of studies reported positive outcomes, including improved patient engagement, increased knowledge, and behavior change. However, the authors noted that most studies were small-scale and short-term, with limited evidence for long-term clinical outcomes.

2.6.3. Large Language Models and Healthcare

The recent emergence of large language models (LLMs) such as GPT-4, Google's Gemini, and Claude has opened new possibilities for healthcare chatbots (Yu et al., 2018). Unlike earlier chatbot approaches that required extensive manual curation of response libraries, LLMs can generate contextually appropriate responses across a wide range of health topics by leveraging vast training corpora of medical literature, clinical guidelines, and health information.

Google's Gemini API, which powers the KWUST SmartHealth chatbot, provides access to Google's Gemini 2.0 Flash model optimized for speed and efficiency (Google LLC, 2024). The model demonstrates strong performance on medical question-answering benchmarks and can handle the types of general health inquiries expected in a university health center context.

However, the use of LLMs in healthcare raises important considerations (Yu et al., 2018; 2018):

- **Accuracy and Hallucination:** LLMs can generate plausible-sounding but incorrect information. The chatbot in SmartHealth includes disclaimers that it is not a substitute for professional medical advice.

- **Privacy:** Patient health information must not be shared with external AI services. SmartHealth addresses this by not storing chat history on the server and not sending patient-identifiable information to the Gemini API.
- **Bias:** LLMs may exhibit biases present in their training data, potentially leading to differential responses across demographic groups.
- **Regulatory Compliance:** Healthcare AI applications must comply with relevant regulations regarding medical device classification and data protection.

KWUST SmartHealth's chatbot implementation addresses these concerns through several design choices. The chatbot is positioned as a general health information tool rather than a diagnostic or triage system. Rate limiting prevents API abuse, and the no-history-storage policy protects user privacy. The floating widget interface provides easy access while maintaining separation from clinical data.

2.7. Patient Portals and Personal Health Records

2.7.1. Patient Engagement Through Digital Portals

Patient portals are secure web-based or mobile applications that allow patients to access their health information, communicate with healthcare providers, and manage aspects of their care digitally (Anderson & Agarwal, 2011; Irizarry et al., 2015). The concept of patient portals emerged from the broader movement toward patient-centered care and shared decision-making, recognizing that informed and engaged patients achieve better health outcomes.

The distinction between patient portals and personal health records (PHRs) is important: patient portals are typically tethered to a specific healthcare provider's EHR system, while PHRs are patient-controlled repositories of health information that may aggregate data from multiple sources. KWUST SmartHealth implements a patient portal that is integrated with the institutional system, providing students with access to their medical records, prescriptions, and appointment history within the university health center.

2.7.2. Features and Impact of Patient Portals

Research has identified several features of patient portals that contribute to improved health outcomes (Irizarry et al., 2015; 2015):

Feature	Description	Impact

View Medical Records	Access to diagnoses, lab results, medications, and visit summaries	Improved understanding of health status; better preparation for appointments
Secure Messaging	Electronic communication with healthcare providers	Reduced phone call volume; improved access to providers; better continuity of care
Appointment Management	Self-service booking, cancellation, and reminders	Reduced no-show rates; improved patient satisfaction; reduced administrative burden
Medication Management	View current medications; renewal requests; adherence tracking	Improved medication adherence; reduced medication errors
Health Education	Targeted educational materials based on conditions or medications	Increased health literacy; better self-management of chronic conditions
Billing and Payments	View charges, make payments, review insurance claims	Improved financial transparency; reduced billing inquiries

Table 17: Patient Portal Features and Their Impact on Healthcare Outcomes

Loya et al. (Irizarry et al., 2015) conducted a usability analysis of a university health center patient portal and found that while most users could complete basic tasks, usability issues with navigation and information presentation reduced satisfaction and efficiency. This finding underscores the importance of user-centered design in portal development, a principle that guides the SmartHealth implementation.

2.7.3. Adoption Factors and Challenges

Despite the demonstrated benefits, patient portal adoption remains below potential, particularly among certain demographic groups (Irizarry et al., 2015). Factors influencing adoption include:

- **Digital Literacy:** Users with limited experience with web applications may struggle to navigate portal interfaces and may require training or simplified interfaces.
- **Perceived Value:** Patients who do not perceive significant benefit from portal features are less likely to register and use the system.
- **Privacy Concerns:** Concerns about the security of health information can deter adoption, particularly among sensitive populations.
- **Access Barriers:** Lack of reliable internet access or suitable devices can exclude segments of the patient population.
- **Registration Complexity:** Complicated registration processes involving multiple verification steps can discourage initial adoption.

KWUST SmartHealth addresses these challenges through a simplified interface designed for varying levels of technical literacy, role-based views that present only relevant information, responsive design supporting mobile access, and a streamlined registration process with verification by staff.

2.8. Security and Privacy in Health Information Systems

2.8.1. Regulatory Frameworks

Health information is among the most sensitive categories of personal data, and its protection is governed by regulatory frameworks in most countries (Bates & Gawande, 2003). While KWUST operates in a jurisdiction with evolving data protection laws, international frameworks guide best practices:

Framework	Scope	Key Requirements
HIPAA (US)	Health plans, healthcare providers, and clearinghouses	Privacy Rule (patient rights over health data); Security Rule (administrative, physical, technical safeguards); Breach Notification Rule
GDPR (EU)	All organizations processing EU resident data	Data minimization; consent; right to access, rectification, erasure; data portability; breach notification within 72 hours
ISO 27799	Health informatics security management	Implementation guidance for ISO/IEC 27002 in healthcare; risk assessment; access control; audit logging

Table 18: Major Regulatory Frameworks for Health Information Security

While KWUST SmartHealth is not directly subject to HIPAA or GDPR, these frameworks provide valuable guidance for implementing appropriate security controls. The system's security architecture incorporates principles aligned with

these standards, including role-based access control, audit logging, data encryption, and user consent mechanisms.

2.8.2. Technical Security Measures

Healthcare information systems must implement multiple layers of security controls to protect sensitive data (Bates & Gawande, 2003; 2003):

- **Authentication:** Verifying the identity of users before granting system access. KWUST SmartHealth uses JWT-based authentication with bcrypt password hashing, providing strong protection against credential compromise.
- **Authorization:** Ensuring that authenticated users can only access data and perform actions appropriate to their role. The system enforces role-based access control (RBAC) at the middleware level, with separate interfaces and API endpoints for students, staff, and administrators.
- **Encryption:** Protecting data both in transit (via HTTPS/TLS for all client-server communication) and at rest (via PostgreSQL connection encryption and database-level access controls).
- **Audit Logging:** Recording all user actions with timestamps, user identifiers, and action types to enable accountability and forensic analysis in the event of a security incident.
- **Input Validation:** Sanitizing all user inputs to prevent injection attacks, with Content Security Policy (CSP) headers to mitigate cross-site scripting (XSS) risks.
- **Session Management:** Implementing token expiry and secure cookie handling to prevent session hijacking.

2.8.3. Privacy Concerns and Trust

Patient trust is essential for the successful adoption of health information systems. Studies have shown that privacy concerns are among the most significant barriers to patient portal adoption and health data sharing (Anderson & Agarwal, 2011; Irizarry et al., 2015). Patients need assurance that their health information will be:

- **Confidential:** Accessible only to authorized individuals for legitimate purposes
- **Secure:** Protected against unauthorized access, breaches, and data loss
- **Accurate:** Correctly recorded and maintained
- **Accessible:** Available to the patient for review and correction

KWUST SmartHealth addresses these concerns through transparent data handling policies communicated during registration, role-based access controls that prevent unauthorized viewing of medical records, audit logs that document all access to sensitive data, and patient portal access that allows students to review their own records at any time.

2.9. Comparison of Existing Approaches

Existing healthcare management solutions vary significantly in their scope, cost, and suitability for university health centers. The table below provides a comprehensive

comparative overview of existing approaches across key evaluation criteria relevant to the KWUST context.

	Appointment. Scheduling	**Medical Records**	**Medicine Inventory**	**AI Chatbot**	**Patient Portal**	**Admin Dashboard**	**Cos**
Manual / Paper-based	—	—	—	—	—	—	Low
Enterprise EHR (Epic, Cerner, OpenMRS)	✓	✓	✓	—	✓	✓	High
Univ. Clinic Systems (O'Brien, Alshazly)	✓	✓	—	—	—	Partial	Medium
Appointment Platforms (Zocdoc, Practo)	✓	—	—	—	—	—	Medium
Inventory Systems (Singh)	—	—	✓	—	—	Partial	Low
Patient Portals Only	—	Partial	—	—	✓	—	Medium
Standalone Chatbots	—	—	—	✓	—	—	Low
KWUST SmartHealth	✓	✓	✓	✓	✓	✓	**Low**

Table 19: Comprehensive Comparison of Existing Healthcare Management Approaches

Enterprise EHR systems such as Epic, Cerner, and OpenMRS offer comprehensive functionality for large healthcare organizations (Menachemi & Collum, 2011). These systems provide integrated clinical data management, order entry, results reporting, billing, and patient portal features. However, their licensing costs, implementation complexity, and infrastructure requirements make them impractical for small to medium-sized university health centers (Kruse et al., 2016). A typical Epic implementation costs millions of dollars and requires months of planning and configuration.

University-specific clinic management systems address many institutional needs at a lower cost point than enterprise EHRs. These systems typically include appointment scheduling, patient registration, and basic medical record keeping. However, they often lack medicine inventory management, AI-powered features, and comprehensive administrative dashboards, requiring integration with additional systems to achieve full functionality.

Standalone appointment scheduling platforms like Zocdoc and Practo provide user-friendly booking experiences and have demonstrated the value of digital scheduling. However, these platforms operate independently from clinical data systems, meaning that appointment data does not flow into medical records and prescribing workflows. This separation creates data silos that undermine the goal of comprehensive, integrated healthcare delivery.

Medicine inventory management systems offer robust stock tracking and ordering capabilities but often do not integrate with patient records or clinical workflows (Volland et al., 2017). Stock levels must be updated independently of actual medication dispensing, creating opportunities for errors and inconsistencies.

Patient portals are valuable for engaging patients in their care, but most are designed as extensions of existing EHR systems rather than standalone solutions (Irizarry et al., 2015). Institutions without an EHR system often face challenges in deploying a patient portal without also adopting the underlying clinical system.

AI-powered healthcare chatbots are increasingly common but are typically deployed as standalone applications rather than integrated components of broader healthcare management platforms (Lee & Yoon, 2021; Yu et al., 2018). This limits their effectiveness, as they cannot access contextual information such as the patient's medical history or current medications.

2.10. Gaps Identified

Based on the comprehensive review of existing literature and systems presented in this chapter, several significant gaps have been identified:

1. **Lack of Integrated Solutions:** Most existing solutions are designed as standalone systems, addressing specific functions such as appointment scheduling, medical records, or inventory management individually. No widely available solution provides an integrated platform combining all these functions specifically for university health centers. The absence of comprehensive integration means that institutions must either deploy multiple disconnected systems (creating data silos

and administrative overhead) or accept the limitations of a single-function solution.

2. **Prohibitive Cost and Complexity:** Enterprise-level EHR systems offer comprehensive functionality but are prohibitively expensive for small institutions. Licensing fees, implementation costs, and ongoing maintenance expenses can exceed the annual operating budget of a university health center (Kruse et al., 2016). Even open-source alternatives require significant technical expertise for deployment, customization, and maintenance (Chaudhry et al., 2006).

3. **Limited AI Integration:** Few healthcare management systems for small to medium-sized institutions integrate AI-powered capabilities. While large healthcare systems are increasingly adopting AI for clinical decision support, predictive analytics, and patient engagement, these technologies remain out of reach for most university health centers (Shaw et al., 2019; Topol, 2019).

4. **Insufficient Patient Engagement:** Many existing systems do not provide patients with direct, convenient access to their medical records or the ability to actively participate in managing their healthcare. Patient portals remain underutilized, particularly in university settings (Irizarry et al., 2015; 2015).

5. **Inadequate Administrative Oversight:** Healthcare staff and administrators require comprehensive tools for system oversight, user management, audit logging, and reporting. Many existing solutions lack these features or provide them in limited form.

6. **Scalability and Flexibility Constraints:** Existing solutions often struggle to scale with growing student populations and changing institutional requirements. Rigid architectures and vendor locking limit the ability to adapt systems to evolving needs.

7. **Lack of Focus on University Health Centers:** The specific operational context of university health centers serving a young adult population with distinct healthcare needs, operating within an educational institution, and serving a defined community has not been adequately addressed by existing healthcare management solutions.

2.11. How This Project Addresses the Gaps

KWUST SmartHealth addresses the identified gaps through a comprehensive set of design decisions and architectural choices. The table below maps each identified gap to the corresponding solution implemented in the proposed system, demonstrating the direct relationship between the literature review findings and the system design.

Gap Identified	How SmartHealth Addresses It
Lack of Integrated Solutions	Single, unified platform combining appointment scheduling, electronic medical records, prescription management, medicine inventory tracking, AI chatbot, patient portal, and administrative dashboard. All modules share a common database and authentication system, ensuring data consistency and eliminating silos.
Prohibitive Cost and Complexity	Open-source technology stack (Node.js, Express, Directus CMS, PostgreSQL) (Monospace Inc., 2024) eliminates licensing costs. Modular architecture allows phased deployment. Low infrastructure requirements enable deployment on affordable cloud or on-premises servers. Estimated total deployment cost is a fraction of enterprise EHR alternatives.
Limited AI Integration	Integration with Google's Gemini API (Google LLC, 2024) provides a generative AI chatbot capable of answering general health inquiries. Features include rate limit handling, privacy-preserving no-history-storage design, and clear disclaimers about AI limitations. The chatbot is accessible via a floating widget across all system pages.
Insufficient Patient Engagement	Comprehensive patient portal providing students with access to full medical history, prescribed medicines with dosage details, appointment management (book, view, cancel, reschedule), and health reminders. Designed with user-centered principles and responsive layout for mobile access (Irizarry et al., 2015).
Inadequate Administrative Oversight	Dedicated admin dashboard with user management (create, verify, suspend, delete accounts across all roles), audit logs documenting all user actions with timestamps and user IDs, medical records overview across all patients, and medicines overview with stock level and expiry monitoring.

Scalability and Flexibility Constraints	PostgreSQL backend provides robust data management for growing datasets. Stateless Node.js API supports horizontal scaling by adding application server instances. Directus headless CMS enables flexible content modeling without code changes, allowing the system to adapt to evolving requirements. Modular architecture allows individual components to be updated or replaced independently.
Lack of Focus on University Health Centers	System designed specifically for university health center workflows based on consultations with KWUST students and staff. Features tailored to the student population (role-based access, academic calendar considerations) and institutional context (campus-based deployment, integration with university systems).

Table 20: Mapping of Identified Gaps to Proposed KWUST SmartHealth Solutions

The system architecture follows a three-tier design as illustrated below. The presentation layer consists of EJS templates and client-side JavaScript rendered by the browser, providing responsive layouts with dark mode support. The application layer comprises Node.js and Express, handling HTTP routing, JWT-based authentication, session management, role-based access control, and PDF generation via PDFKit. The data layer uses Directus as a headless Content Management System (CMS) with PostgreSQL as the database engine, providing a RESTful API for all data operations.

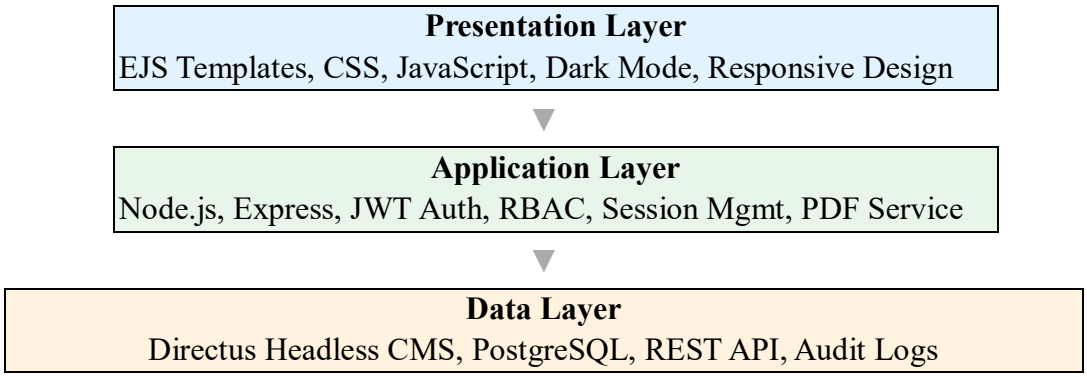


Figure 1: Three-Tier System Architecture of KWUST SmartHealth

The integration of the Gemini-powered AI chatbot provides students with instant answers to general health questions, reducing the burden on healthcare staff while promoting health literacy (Google LLC, 2024). The user-centered design approach ensures the system is accessible and intuitive for all user groups, including those with limited technical literacy (Irizarry et al., 2015). The comprehensive patient portal empowers students to access their medical history, view prescribed medicines, and

manage appointments; features that existing solutions provide only in isolation or not at all.

A distinguishing feature of the proposed system is the deep integration between modules. When a healthcare staff member creates a medical record and prescribes medicines, the inventory module automatically updates stock levels. The patient portal immediately reflects the new record and prescription. The administrative dashboard captures the action in audit logs. This seamless integration eliminates the data silos that plague standalone solutions and provides a unified view of healthcare delivery at the institutional level.

The architecture also supports the PDF generation pipeline, which enables printing of medical records and prescriptions with institutional branding. This feature bridges the gap between the digital and physical worlds, accommodating scenarios where paper records are still required for external referrals, insurance claims, or patient preference.

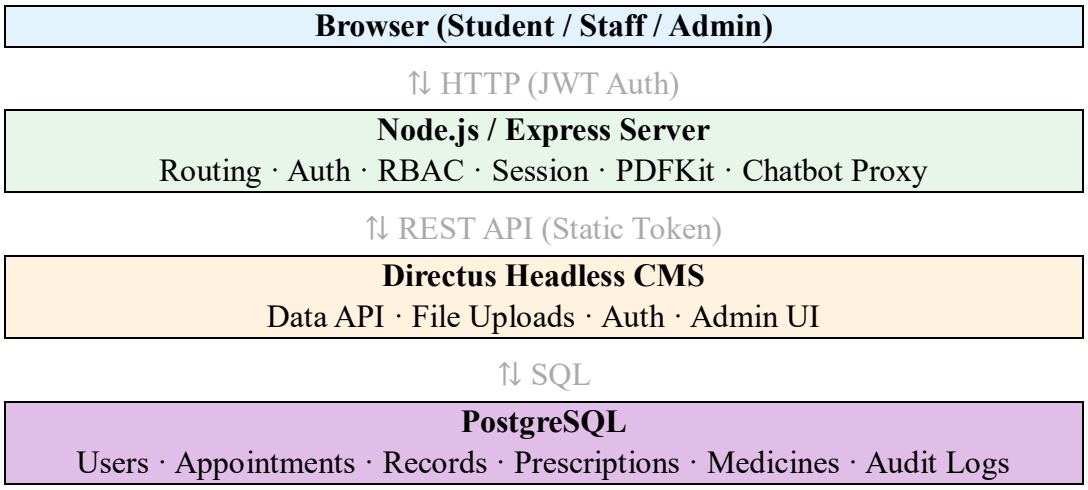


Figure 2: Detailed Layered Architecture of KWUST SmartHealth Showing Data Flow Between Components

2.12. Summary

This chapter has provided a comprehensive review of literature and existing systems related to healthcare management, with a specific focus on the university health center context. The theoretical framework grounding this research encompasses the DeLone and McLean IS Success Model for evaluating system effectiveness, the Technology Acceptance Model for understanding user adoption, user-centered design principles for guiding development, and health informatics theory for ensuring clinical relevance.

The review of electronic health record systems revealed that while these systems offer substantial benefits in terms of clinical efficiency, patient safety, and care coordination, their high cost and implementation complexity limit accessibility for small institutions like university health centers. Appointment scheduling systems have demonstrated improvements in patient satisfaction and operational efficiency, but standalone scheduling platforms lack integration with clinical data. Medicine

inventory management is critical for healthcare delivery, and digital systems can significantly reduce stockouts and waste, yet these systems typically operate in isolation from patient records and prescribing workflows.

The emerging field of AI in healthcare, particularly conversational AI, offers new opportunities for patient engagement and health education. However, the integration of AI capabilities into comprehensive healthcare management platforms for small institutions remains limited. Patient portals improve patient engagement and health outcomes but are typically tied to expensive EHR systems.

The comparative analysis of existing approaches confirmed that no widely available solution provides an integrated, affordable platform combining appointment scheduling, electronic medical records, prescription management, medicine inventory tracking, AI chatbot assistance, patient portal access, and administrative oversight specifically designed for university health centers. The identified gaps- lack of integration, prohibitive cost, limited AI features, insufficient patient engagement, and scalability constraints- provide the motivation and justification for the KWUST SmartHealth system.

By addressing these gaps through an integrated, open-source, user-centered platform, KWUST SmartHealth aims to transform healthcare delivery at Kiriri Women's University of Science and Technology and provide a replicable model for other university health centers facing similar challenges (Adler-Milstein & Bates, 2010; World Health Organization, 2021).

3. CHAPTER THREE: METHODOLOGY

3.1. Introduction

This chapter describes the technical approach and development methodology for implementing the proposed KWUST SmartHealth system. The chapter is organized as follows: Section 3.2 presents the overall system architecture. Section 3.3 details the technology stack and justification for each choice. Section 3.4 explains the development methodology (Agile/Scrum). Section 3.5 describes the key modules and their functionalities. Section 3.6 provides a detailed explanation of the PDF generation pipeline. Section 3.7 outlines the evaluation metrics and testing procedures. Section 3.8 addresses ethical considerations related to health data handling.

3.2. System Architecture

The system follows a three-tier client-server architecture with three primary layers communicating via RESTful APIs and a headless CMS. The architecture is designed to be modular, scalable, and secure.

3.2.1. High-Level Architecture Diagram

The system follows a four-tier architecture comprising the presentation layer, application layer, CMS layer, and data layer, with an external AI service. Figure 3.1 illustrates the system components and the interactions between them.

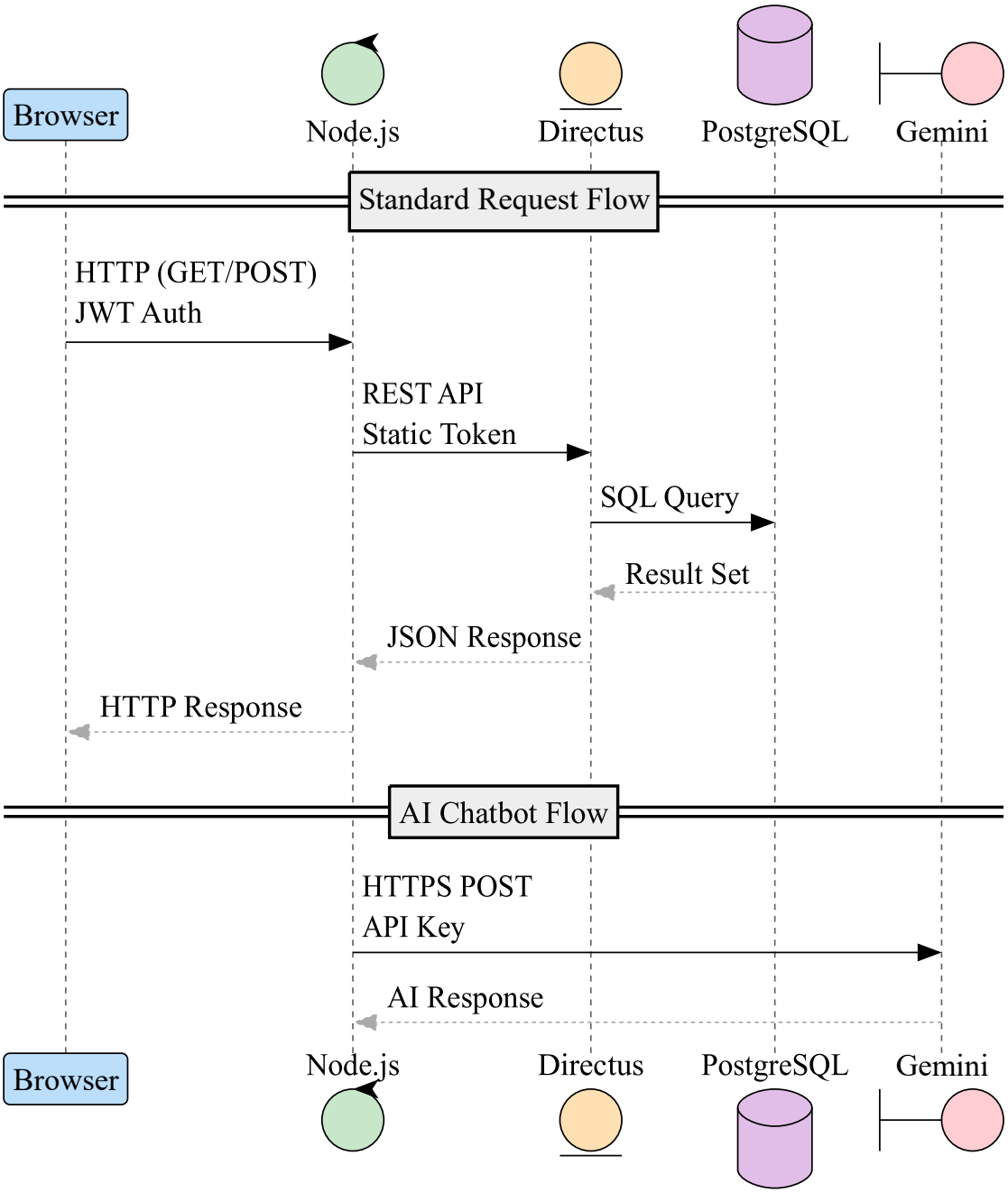


Figure 3: System Architecture Sequence Diagram Showing Inter-Component Communication

3.2.2. Component Descriptions

Component	Role	Deployment
Browser (Student, Staff, Admin)	Renders EJS templates; handles client-side interactions; manages local state	Any modern web browser (Chrome, Firefox, Safari)
Node.js / Express Server	Routes HTTP requests; manages sessions; enforces role-based access; generates PDFs	Cloud or on-premises server (Node.js runtime)
Directus Headless CMS	Provides RESTful API for database operations; manages file uploads; handles authentication	Same or separate server as Node.js
PostgreSQL Database	Stores users, appointments, medical records, prescriptions, medicines, and audit logs	Database server (managed or self-hosted)
Gemini API (External)	AI chatbot service for health inquiries	Google Cloud (external API)

Table 21: System Components and Their Roles

3.2.3. Data Flow (Appointment Booking)

The data flow for a typical appointment booking scenario proceeds as follows:

1. Student logs into the system via the browser and navigates to the appointment page.
2. Browser sends a GET request to the Node.js server to fetch available staff and time slots.
3. Server queries Directus for staff schedules and available appointments; Directus queries PostgreSQL.
4. Server returns available slots to the browser, rendered as a calendar or list.
5. Student selects a staff member, date, and time slot, then submits the booking.
6. Browser sends a POST request with appointment details (student ID, staff ID, date, time).
7. Server validates the request (slot availability, no double-booking), then creates a record via the Directus API.
8. Directus inserts the appointment record into PostgreSQL and returns the created record.
9. Server sends a confirmation response to the browser; student sees a success message.
10. Server optionally triggers a notification (reminder) for the upcoming appointment.

3.3. Technology Stack

3.3.1. Backend Technologies

Technology	Version	Purpose
Node.js	20 LTS	JavaScript runtime for server-side logic
Express	4.x	Web framework for routing, middleware, session management
PDFKit	0.15	PDF generation for medical records and prescriptions
node-fetch / axios	—	HTTP client for Gemini API requests
bcrypt	—	Password hashing for authentication
jsonwebtoken	—	JWT generation and verification

Table 22: Backend Software Stack

3.3.2. Frontend Technologies

Technology	Version	Purpose
HTML5 / EJS	—	Template rendering with embedded JavaScript
CSS3	—	Styling, responsive layout, dark mode support
JavaScript (Vanilla)	ES6	Client-side interactivity, form validation, API calls
Fetch API	—	HTTP requests to backend (no external library)
Chart.js	4.x	Charts for dashboard analytics

Table 23: Frontend Software Stack

3.3.3. Database and CMS

Technology	Version	Purpose
Directus	10.x	Headless CMS providing REST API, authentication, file management, admin UI
PostgreSQL	15+	Relational database for all application data
PostGIS (optional)	—	Geospatial extensions (future use for location-based services)

3.4. Development Methodology

3.4.1. Approach: Agile (Scrum)

The project will follow an Agile Scrum methodology with 2-week sprints. This approach is appropriate for a web application development project where requirements may evolve during implementation, particularly for the user interface and AI chatbot integration, which require iterative refinement.

Justification for Agile:

- The user interface design may require multiple iterations based on feedback from students and staff.
- The AI chatbot integration with the Gemini API requires iterative testing for prompt engineering and error handling.
- Early user feedback (from testing with fellow students and staff) can inform usability improvements.
- Risk mitigation: Core workflows (appointment booking, medical records) can be demonstrated early (sprint 3), leaving time for refinement.

3.4.2. Sprint Structure (6 sprints, 12 weeks total)

Sprint	Weeks	Activities	Deliverable
Sprint 0	Week 0	Environment setup: install Node.js, Directus, PostgreSQL; configure project structure; set up version control	Working development environment with Directus and PostgreSQL connected
Sprint 1	Weeks 1–2	User authentication: registration, login, role-based access (student, staff, admin); database schema de-sign for users and roles	Login and registration working for all three roles
Sprint 2	Weeks 3–4	Appointment scheduling: staff defines time slots; student books, views, cancels appointments; calendar UI	Functional appointment booking flow

Sprint 3	Weeks 5–6	Medical records module: staff creates records with diagnoses and prescriptions; patient portal to view history; medicine inventory CRUD	Medical records and inventory management working
Sprint 4	Weeks 7–8	PDF generation service; AI chatbot integration with Gemini API; admin dashboard with user management	PDF download, chatbot, and admin panel functional
Sprint 5	Weeks 9–10	Full integration testing; dark mode; responsive design polish; reminder notifications; search and filtering	Complete system with all features integrated
Sprint 6	Weeks 11–12	User acceptance testing; performance benchmarking; bug fixes; documentation	Final deliverable with metrics report

Table 25: Sprint Plan for KWUST SmartHealth Development

3.4.3. Daily and Weekly Practices

- **Daily standup** (15 minutes, self-tracked): What was done yesterday? What is today's goal? Any blockers?
- **Weekly sprint review** (Friday): Demonstrate working features to supervisor.
- **Sprint retrospective** (end of each sprint): What went well? What needs improvement?
- **Version control**: Git with commit messages following the conventional commits format.

3.5. Key Modules and Functionalities

3.5.1. Module 1: User Management and Authentication

Function	Description
Student registration	Student creates account with name, email, student ID, password. Account requires staff/admin verification before first login.
Staff registration	Staff registers with name, email, department, password. Role assigned as healthcare staff
Admin registration	Initial admin account created by system developer via Directus CMS dashboard; additional admins managed through admin panel. Full system access
Login and session	JWT-based authentication; session persists via token stored in local storage or cookie; role-based redirect after login
Password reset	Admin can reset user passwords; future integration with email-based reset flow.

Table 26: User Management Module Functions

3.5.2. Module 2: Appointment Scheduling

Function	Description
Staff defines time slots.	Staff creates available appointment slots with date, start time, end time, and maximum patients per slot.
Student books appointment	Student views available slots, selects a time, and confirms booking.
View appointments	Students see their upcoming and past appointments; staff see all appointments for their slots.
Cancel/reschedule	Students can cancel a booking; staff can mark appointments as completed or cancelled.
Appointment reminders	System displays reminders on dashboard for upcoming appointments; future SMS/email integration.

Table 27: Appointment Scheduling Module Functions

3.5.3. Module 3: Medical Records and Prescriptions

Function	Description
Create medical record	Staff creates a record for a student visit: diagnosis, symptoms, notes, and prescribed medicines.
Prescription management	Staff selects medicines from inventory and sets the dosage, duration, and instructions for each prescription.
View medical history	Student views their complete medical history including diagnoses, prescriptions, and visit dates.
Search and filter	Staff can search records by student name, date range, or diagnosis.
PDF export	Students and staff can download medical records or prescriptions as a PDF with institutional logo and timestamp.

Table 28: Medical Records Module Functions

3.5.4. Module 4: Medicine Inventory Management

Function	Description
Add medicine	Staff adds new medicine to the inventory with the name, category, quantity, and expiry date.
Search and filter	Search medicines by name or category; filter by stock level or expiry status
Stock tracking	Quantities update automatically when medicines are prescribed; low-stock alerts for quantities below threshold.
Category management	Medicines grouped by therapeutic category (pain relief, antibiotics, etc.) for easier navigation

Table 29: Medicine Inventory Module Functions

3.5.5. Module 5: AI Chatbot

Feature	Description
Floating widget	Chatbot accessible via a floating action button on all pages; slides up as a chat panel.
Gemini API integration	Messages sent to Google Gemini 2.0 Flash model; responses rendered as bot messages
Rate limit handling	429 errors display a cooldown countdown to the user; retry-after time parsed from API response
No history storage	Each request is standalone; no chat history stored on the server for privacy.

Table 30: AI Chatbot Module Features

3.5.6. Module 6: Admin Dashboard and Reporting

Function	Description
User management	View, verify, suspend, or delete user accounts across all roles.
Audit logs	System logs showing user actions (login, record creation, appointment changes) for accountability.
Medical records overview	View all medical records across patients; filter by staff, date, or diagnosis.
Medicines overview	View all medicines; monitor stock levels and expiry dates across the entire inventory.

Table 31: Admin Dashboard Module Functions

3.6. PDF Generation Pipeline

3.6.1. Overview

The PDF generation pipeline enables printing of medical records and prescriptions. It uses the PDFKit library in Node.js to produce documents styled with the institutional brand. The pipeline consists of three phases: document initialization, content generation, and output delivery.

3.6.2. Phase 1: Document Initialization

Step	Operation	Method
1	Create PDF document	new PDFDocument() with A4 page size, margins configured
2	Add institutional header	Blue header bar (60px height) with KWUST logo at top-left and document title right-aligned
3	Add timestamp	Right-aligned below header bar: "Generated: {date} {time}" in italic gray
4	Begin content section	Draw horizontal separator line below timestamp

Table 32: PDF Document Initialization Steps

3.6.3. Phase 2: Content Generation

For medical records, the pipeline writes patient details (name, date, staff), then iterates over diagnoses, symptoms, notes, and prescribed medicines. For prescriptions, it creates a focused document listing only the medicines with dosage, duration, and instructions.

Step	Operation	Method
1	Write patient header	Bold patient name, visit date, staff name
2	Write diagnosis section	Bulleted list of diagnoses with symptoms and notes
3	Write prescribed medicines	Numbered list: medicine name, dosage, duration with "days" suffix, instructions
4	Handle multi-page	Automatic page break with header repeated on each new page

Table 33: PDF Content Generation Steps

3.6.4. Phase 3: Output

Step	Operation	Method
1	Finalize document	doc.end(): flushes the PDF stream
2	Stream to response	Pipe PDF buffer to HTTP response with content-type application/pdf
3	Browser handling	Browser displays PDF inline or triggers download based on Content-Disposition header.

Table 34: PDF Output Delivery

Steps Code pseudocode for the PDF generation:

```
def generate_medical_record(student, staff, diagnosis, symptoms, notes, medicines):  doc = PDFDocument()  add_header(doc, "Medical Record")  add_timestamp(doc)  doc.write("Patient: " + student.name)  doc.write("Date: " + visit.date)  doc.write("Diagnosis: " + diagnosis)  doc.write("Symptoms: " + symptoms)  doc.write("Notes: " + notes)  doc.write("Prescribed Medicines:")  for med in medicines:  doc.write(str(i) + ". " + med.name + " - " + med.dosage + " x " + med.duration + " days")  if med.instructions:  doc.write("  Instructions: " + med.instructions)  return doc.output()
```

3.7. Evaluation Metrics and Testing

3.7.1. Performance Metrics

Metric	Target	Measurement Method	Test Environment
Page load time (dashboard)	< 2 seconds	Chrome DevTools Network tab — DOMContentLoaded time	5 loads, averaged across different user roles
Appointment booking time	< 3 seconds	Time from clicking "Book" to confirmation displayed	20 booking attempts, averaged
PDF generation time	< 5 seconds	Time from clicking "Download" to file receipt	15 generations (mix of records and prescriptions)
Chatbot response time	< 5 seconds	Time from sending message to receiving bot reply	20 queries (including Gemini API latency)
Search response time	< 2 seconds	Time from typing search term to results displayed	20 searches across different modules
API endpoint latency	< 500 ms	Server-side logging of request-to-response duration	50 requests across all major endpoints
System uptime	> 99%	Monitor during testing period	Continuous monitoring during 2-week test period

Table 35: Performance Metrics and Targets

3.7.2. Testing Procedures Test 1:

Functional Testing (All Objectives)

- **Setup:** Deployed system with test data (20 sample students, 3 staff, 1 admin, 30 medicines, 50 appointments).
- **Procedure:** Execute each user story (appointment booking, record creation, prescription, inventory update, chatbot query, PDF download, admin verification). Verify correct behavior and error handling.
- **Success criteria:** All user stories pass without errors.

Test 2: Usability Testing (Objective 8)

- **Setup:** 10 volunteer participants (5 students, 3 staff, 2 administrators). Each receives a 5-minute orientation.

- **Tasks:** Book an appointment, view medical history, create a medical record (staff), add a medicine (staff), download a PDF, ask the chatbot a question.
- **Measurement:** Task completion rate, time per task, number of errors.
- **Survey:** System Usability Scale (SUS) questionnaire after testing.
- **Success criteria:** SUS score > 70; task completion rate > 90%.

Test 3: Security Testing

- **Setup:** Test accounts for each role.
- **Procedure:** Attempt unauthorized access (student tries staff endpoints, unauthenticated user tries protected routes, SQL injection attempts on search fields).
- **Success criteria:** All unauthorized requests return 401/403; no SQL injection succeeds; JWT tokens expire correctly.

Test 4: Responsive Design and Cross-Browser Testing

- **Setup:** Chrome, Firefox, Safari browsers; desktop (1920×1080) and mobile (375×667) viewports.
- **Procedure:** Navigate through all pages; verify layout, form inputs, and data tables render correctly.
- **Success criteria:** No layout breaks; all functionality works on all tested browsers and viewports.

3.7.3. Pass/Fail Criteria

The project will be considered successful if:

Criterion	Threshold
All core features functional (appointments, records, inventory, chatbot, PDF, admin)	100%
Page load time	< 2 seconds (mean)
PDF generation time	< 5 seconds (mean)
Chatbot response time	< 5 seconds (mean)
Usability (SUS score)	> 70 / 100
Task completion rate	> 90%
Unauthorized access blocked	100% of attempts rejected
No unrecoverable crashes	0 critical bugs

Table 36: Pass/Fail Criteria for Project Success

3.8. Ethical Considerations

3.8.1. Data Privacy and Security

Health data is sensitive and requires special protection. The system implements the following measures:

Measure	Implementation
Authentication	JWT-based authentication with role-based access control; passwords hashed with bcrypt
Authorization	Middleware-enforced role checks on all routes; students cannot access staff/admin endpoints
Data encryption in transit	HTTPS for all client-server communication; API calls over TLS
Data encryption at rest	PostgreSQL connection encryption; database-level access controls
Session management	Token expiry after configurable period; secure HTTP-only cookies where applicable
Audit logging	All user actions logged with timestamp, user ID, and action type for accountability.
Input sanitization	Content Security Policy (CSP) headers to prevent XSS attacks; input validation on all forms

Table 37: Data Privacy and Security Measures

3.8.2. User Consent and Transparency

- Users (students, staff, administrators) must consent to data collection during account registration.
- The system clearly indicates what data is collected and for what purpose.
- Students can view their own medical records and prescription history at any time.
- The AI chatbot provides general health information only and explicitly states it is not a substitute for professional medical advice.
- Chat history is not stored on the server, ensuring user privacy.

3.8.3. Accessibility and Fairness

- The interface is designed to accommodate users with varying levels of technical literacy.
- Dark mode support reduces eye strain for users who prefer it.
- Role-based views ensure users only see data relevant to their responsibilities.
- The system uses responsive web design to work across desktop and mobile devices, improving accessibility for users who may not have access to a computer.

3.9. Summary

This chapter presented the methodology for developing the KWUST SmartHealth healthcare management system. The system uses a three-tier web architecture with Node.js and Express on the application layer, Directus as a headless CMS, and PostgreSQL as the database. An Agile Scrum methodology with 2-week sprints will guide development over 12 weeks. Key modules include user management, appointment scheduling, medical records with prescriptions, medicine inventory, AI chatbot, PDF generation, and an admin dashboard. Evaluation metrics target sub-2-second page loads, sub-5-second PDF generation, a SUS usability score above 70, and 100% unauthorized access rejection. Ethical considerations include encryption of health data, role-based access control, user consent, and transparency about AI chatbot limitations.

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